Advanced Proof Patterns

Example 1 : Proof by Cases (Three - Way Split)

Prove: For all integers n, $n < 0 \vee n = 0 \vee n > 0$

$$\frac{}{n < 0 \vee n = 0 \vee n > 0}\ [\text{trichotomy of integers, axiom}]$$

This uses the trichotomy property as an axiom. To prove something about all integers, we can case-analyze on these three possibilities.

Example 2 : Multi - Level Case Analysis

Prove: $(p \vee q) \wedge (r \vee s) \Rightarrow (p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)$

$$\cfrac{\cfrac{\cfrac{\cfrac{}{p \wedge r}\ [\wedge\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}\ [\vee\text{-intro}] \quad \cfrac{\cfrac{}{p \wedge s}\ [\wedge\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}\ [\vee\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}\ [\vee \text{ elim on r} \vee \text{s}] \quad \cfrac{\cfrac{\cfrac{}{q \wedge r}\ [\wedge\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}\ [\vee\text{-intro}] \quad \cfrac{\cfrac{}{q \wedge s}\ [\wedge\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}\ [\vee\text{-intro}]}{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}\ [\vee \text{ elim on r} \vee \text{s}]}{\cfrac{(p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}{((p \vee q) \wedge (r \vee s)) \Rightarrow (p \wedge r) \vee (p \wedge s) \vee (q \wedge r) \vee (q \wedge s)}\ [\Rightarrow\text{-intro}^{[1]}]}\ [\vee \text{ elim on p} \vee \text{q}]$$

Nested case analysis on two disjunctions, exploring all four combinations.

Example 3 : Proof by Mathematical Induction (Base and Step)

Prove: For all $n \in \mathbb{N}$, sum(1 to n) = n(n+1)/2

Base case ($n = 0$):

$$\cfrac{\cfrac{\cfrac{\cfrac{\ulcorner \mathit{true} \urcorner^{[1]}}{\mathit{sum\_to}(0) = 0}\ [\text{definition}]}{0 * (0+1)\ \mathsf{div}\ 2 = 0}\ [\text{arithmetic}]}{\mathit{sum\_to}(0) = 0 * (0+1)\ \mathsf{div}\ 2}\ [\text{equality}]}{\mathit{true} \Rightarrow (\mathit{sum\_to}(0) = 0 * (0+1)\ \mathsf{div}\ 2)}\ [\Rightarrow\text{-intro}^{[1]}]$$

Inductive step (assume for n, prove for n+1):

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{sum\_to}(n) = n * (n+1) \;\mathsf{div}\; 2 \urcorner^{[1]}}{\mathit{sum\_to}(n+1) = \mathit{sum\_to}(n) + (n+1)}\;[\text{definition}]}{\mathit{sum\_to}(n+1) = n * (n+1) \;\mathsf{div}\; 2 + (n+1)}\;[\text{substitution}]}{\mathit{sum\_to}(n+1) = (n * (n+1) + 2 * (n+1)) \;\mathsf{div}\; 2}\;[\text{algebra}]}{\mathit{sum\_to}(n+1) = (n+1) * (n+2) \;\mathsf{div}\; 2}\;[\text{factoring}]}{(\mathit{sum\_to}(n) = n * (n+1) \;\mathsf{div}\; 2) \Rightarrow (\mathit{sum\_to}(n+1) = (n+1) * (n+2) \;\mathsf{div}\; 2)}\;[\Rightarrow\text{-intro}^{[1]}]$$

By induction, the formula holds for all natural numbers.

Example 4 : Structural Induction on Lists

Prove: For all sequences s, reverse(reverse(s)) = s

Base case (empty sequence):

$$\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{true} \urcorner^{[1]}}{\mathit{reverse}(\mathit{emptyseq}) = \mathit{emptyseq}}\;[\text{definition}]}{\mathit{reverse}(\mathit{reverse}(\mathit{emptyseq})) = \mathit{reverse}(\mathit{emptyseq})}\;[\text{substitution}]}{\mathit{reverse}(\mathit{reverse}(\mathit{emptyseq})) = \mathit{emptyseq}}\;[\text{definition}]}{\mathit{true} \Rightarrow (\mathit{reverse}(\mathit{reverse}(\mathit{emptyseq})) = \mathit{emptyseq})}\;[\Rightarrow\text{-intro}^{[1]}]$$

Inductive step (assume for s, prove for cons(x, s)):

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{reverse}(\mathit{reverse}(s)) = s \urcorner^{[1]}}{\mathit{reverse}(\mathit{cons}(x, s)) = \mathit{append}(\mathit{reverse}(s), x)}\;[\text{definition}]}{\mathit{reverse}(\mathit{reverse}(\mathit{cons}(x, s))) = \mathit{reverse}(\mathit{append}(\mathit{reverse}(s), x))}\;[\text{substitution}]}{\mathit{reverse}(\mathit{append}(\mathit{reverse}(s), x)) = \mathit{cons}(x, \mathit{reverse}(\mathit{reverse}(s)))}\;[\text{definition}]}{\mathit{cons}(x, \mathit{reverse}(\mathit{reverse}(s))) = \mathit{cons}(x, s)}\;[\text{inductive hypothesis}]}{(\mathit{reverse}(\mathit{reverse}(s)) = s) \Rightarrow (\mathit{reverse}(\mathit{reverse}(\mathit{cons}(x, s))) = \mathit{cons}(x, s))}\;[\Rightarrow\text{-intro}^{[1]}]$$

By structural induction, reverse(reverse(s)) = s for all sequences.

Example 5 : Constructive Existence Proof

Prove: There exists an even number greater than 10

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{true} \urcorner^{[1]}}{12 = 2 * 6}\;[\text{arithmetic}]}{\mathit{even}(12)}\;[\text{definition of even}, 12 = 2 * 6]}{12 > 10}\;[\text{arithmetic}]}{\mathit{even}(12) \land 12 > 10}\;[\land\text{-intro}]}{\exists\, n : \mathbb{N} \bullet \mathit{even}(n) \land n > 10}\;[\exists\,\text{intro with n = 12}]}{\mathit{true} \Rightarrow (\exists\, n : \mathbb{N} \bullet \mathit{even}(n) \land n > 10)}\;[\Rightarrow\text{-intro}^{[1]}]$$

Constructive proof: we exhibit a specific witness (12).

Example 6 : Non - Constructive Existence Proof

Prove: There exist irrational numbers a and b such that aˆb is rational

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\mathit{irrational}(\mathit{sqrt}(2)) \wedge \mathit{irrational}(\mathit{sqrt}(2)) \wedge \mathit{rational}(\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2)))}\,[\wedge\text{ -intro}]}{\exists\, a, b \bullet \mathit{irrational}(a) \wedge \mathit{irrational}(b) \wedge \mathit{rational}(\mathit{power}(a, b))}\,[\exists\text{intro with a} = \text{b} = \text{sqrt(2)}] \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\mathit{power}(\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2)), \mathit{sqrt}(2)) = \mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2) * \mathit{sqrt}(2))}\,[\text{exponent law}]}{\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2) * \mathit{sqrt}(2)) = \mathit{power}(\mathit{sqrt}(2), 2)}\,[\text{arithmetic}]}{\mathit{power}(\mathit{sqrt}(2), 2) = 2}\,[\text{simplification}]}{\mathit{rational}(2)}\,[\text{known}]}{\mathit{irrational}(\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2))) \wedge \mathit{irrational}(\mathit{sqrt}(2)) \wedge \mathit{rational}(\mathit{power}(\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2)), \mathit{sqrt}(2)))}\,[\wedge\text{ -intro}]}{\exists\, a, b \bullet \mathit{irrational}(a) \wedge \mathit{irrational}(b) \wedge \mathit{rational}(\mathit{power}(a, b))}\,[\exists\text{-intro}]}{\exists\, a, b \bullet \mathit{irrational}(a) \wedge \mathit{irrational}(b) \wedge \mathit{rational}(\mathit{power}(a, b))}\,[\vee\text{ -elim}]}{\mathit{irrational}(\mathit{sqrt}(2)) \Rightarrow (\exists\, a, b \bullet \mathit{irrational}(a) \wedge \mathit{irrational}(b) \wedge \mathit{rational}(\mathit{power}(a, b)))}\,[\Rightarrow\text{ -intro}^{[1]}]$$

Non-constructive: we don't know which case is true, but both lead to the conclusion.

Example 7 : Proof by Strong Induction

Prove: Every natural number $n \geq 2$ has a prime factorization

Base case ($n = 2$):

$$\dfrac{\dfrac{\dfrac{\ulcorner \mathit{true} \urcorner^{[1]}}{\mathit{prime}(2)}\,[\text{definition}]}{\mathit{prime\_factorization}(2)}\,[\text{trivial, singleton factorization}]}{\mathit{true} \Rightarrow \mathit{prime\_factorization}(2)}\,[\Rightarrow\text{ -intro}^{[1]}]$$

Inductive step (assume for all $k < n$, prove for n):

$$\dfrac{\dfrac{\dfrac{}{\mathit{prime\_factorization}(n)}\,[\text{trivial, singleton factorization}] \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\exists\, a, b \bullet 2 \leq a \wedge a < n \wedge 2 \leq b \wedge b < n \wedge n = a * b}\,[\text{definition of composite}]}{\mathit{prime\_factorization}(a)}\,[\text{strong IH, a} < \text{n}]}{\mathit{prime\_factorization}(b)}\,[\text{strong IH, b} < \text{n}]}{\mathit{prime\_factorization}(a * b)}\,[\text{multiplication of factorizations}]}{\mathit{prime\_factorization}(n)}\,[\text{n} = \text{a} * \text{b}]}{\mathit{prime\_factorization}(n)}\,[\vee\text{ -elim}]}{(n \geq 2) \Rightarrow \mathit{prime\_factorization}(n)}\,[\Rightarrow\text{ -intro}^{[1]}]$$

Strong induction: we assume the property for all smaller values, not just n-1.

Example 8 : Proof Using Lemmas

Lemma 1: If n is even, then nˆ2 is even

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{even}(n) \urcorner^{[1]}}{\exists\, k \bullet n = 2 * k}\,[\text{definition of even}]}{\mathit{power}(n, 2) = \mathit{power}(2 * k, 2)}\,[\text{substitution}]}{\mathit{power}(2 * k, 2) = 4 * \mathit{power}(k, 2)}\,[\text{algebra}]}{4 * \mathit{power}(k, 2) = 2 * (2 * \mathit{power}(k, 2))}\,[\text{factoring}]}{\exists\, m \bullet \mathit{power}(n, 2) = 2 * m}\,[\exists\text{intro with m} = 2 * \text{power(k, 2)}]}{\mathit{even}(\mathit{power}(n, 2))}\,[\text{definition of even}]}{\mathit{even}(n) \Rightarrow \mathit{even}(\mathit{power}(n, 2))}\,[\Rightarrow\text{ -intro}^{[1]}]$$

Main theorem: If n^2 is odd, then n is odd

$$
\begin{prooftree}
\AxiomC{}
\RightLabel{[Lemma1]}
\UnaryInfC{$\mathit{even}(\mathit{power}(n, 2))$}
\RightLabel{[contradiction]}
\UnaryInfC{$\mathit{odd}(\mathit{power}(n, 2)) \wedge \mathit{even}(\mathit{power}(n, 2))$}
\RightLabel{[contradiction]}
\UnaryInfC{false}
\RightLabel{[false-elim]}
\UnaryInfC{$\mathit{odd}(n)$}
\AxiomC{}
\RightLabel{[identity]}
\UnaryInfC{$\mathit{odd}(n)$}
\RightLabel{[$\vee$ -elim]}
\BinaryInfC{$\mathit{odd}(n)$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$\mathit{odd}(\mathit{power}(n, 2)) \Rightarrow \mathit{odd}(n)$}
\end{prooftree}
$$

Proof by contrapositive using lemma.

Example 9 : Proof by Minimal Counterexample

Prove: All natural numbers $n \geq 1$ satisfy $p(n)$

$$
\begin{prooftree}
\AxiomC{}
\RightLabel{[base case proved separately]}
\UnaryInfC{$p(1)$}
\RightLabel{[contradiction]}
\UnaryInfC{$\neg\,((p(m))) \wedge p(1)$}
\RightLabel{[contradiction]}
\UnaryInfC{false}
\AxiomC{}
\RightLabel{[minimality of m]}
\UnaryInfC{$\forall\, k \bullet k \geq 1 \wedge k < m \Rightarrow p(k)$}
\RightLabel{[since m − 1 >= 1 ∧ m − 1 < m]}
\UnaryInfC{$p(m - 1)$}
\RightLabel{[by inductive step from p(m − 1)]}
\UnaryInfC{$p(m)$}
\RightLabel{[contradiction]}
\UnaryInfC{$\neg\,((p(m))) \wedge p(m)$}
\RightLabel{[contradiction]}
\UnaryInfC{false}
\RightLabel{[$\vee$ -elim]}
\BinaryInfC{false}
\RightLabel{[$\neg$ -intro$^{[2]}$]}
\UnaryInfC{$\forall\, n \bullet n \geq 1 \Rightarrow p(n)$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$\mathit{true} \Rightarrow (\forall\, n \bullet n \geq 1 \Rightarrow p(n))$}
\end{prooftree}
$$

Minimal counterexample combines well-ordering with contradiction.

Example 10 : Proof by Invariant

Prove: A loop maintains invariant I

Initialization:

$$
\begin{prooftree}
\AxiomC{$\ulcorner \mathit{initial\_state} \urcorner^{[1]}$}
\RightLabel{[verification]}
\UnaryInfC{$\mathit{invariant}(\mathit{initial\_state})$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$\mathit{initial\_state} \Rightarrow \mathit{invariant}(\mathit{initial\_state})$}
\end{prooftree}
$$

Preservation:

$$
\begin{prooftree}
\AxiomC{$\ulcorner \mathit{invariant}(\mathit{before\_state}) \wedge \mathit{executes\_loop\_body} \urcorner^{[1]}$}
\RightLabel{[$\wedge$ -elim-1]}
\UnaryInfC{$\mathit{invariant}(\mathit{before\_state})$}
\RightLabel{[$\wedge$ -elim-2]}
\UnaryInfC{$\mathit{executes\_loop\_body}$}
\RightLabel{[verification]}
\UnaryInfC{$\mathit{invariant}(\mathit{after\_state})$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$(\mathit{invariant}(\mathit{before\_state}) \wedge \mathit{executes\_loop\_body}) \Rightarrow \mathit{invariant}(\mathit{after\_state})$}
\end{prooftree}
$$

Termination:

$$\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{loop\_terminates} \urcorner^{[1]}}{\mathit{invariant}(\mathit{termination\_state})}\;[\text{by preservation}]}{\mathit{invariant}(\mathit{termination\_state}) \wedge \mathit{termination\_condition}}\;[\wedge\text{ -intro}]}{\mathit{desired\_property}}\;[\text{logic}]}{\mathit{loop\_terminates} \Rightarrow \mathit{desired\_property}}\;[\Rightarrow\text{ -intro}^{[1]}]$$

Example 11 : Proof by Diagonalization

Prove: The set of real numbers is uncountable

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{countable}(\mathit{reals}) \urcorner^{[2]}}{\exists f \bullet \mathit{enumeration}(f, \mathit{reals})}\;[\text{definition of countable}]}{\mathit{diagonal\_construction}(r)}\;[\text{diagonal method}]}{\forall n \bullet r \neq \mathit{apply}(f, n)}\;[\text{by construction, differs at nth digit}]}{\mathit{not\_in\_range}(r, f)}\;[\text{previous line}]}{\mathit{not\_in\_range}(r, f) \wedge \mathit{enumeration}(f, \mathit{reals})}\;[\text{contradiction}]}{\mathit{false}}\;[\text{contradiction}]}{\mathit{uncountable}(\mathit{reals})}\;[\neg\text{ -intro}^{[2]}]}{\mathit{true} \Rightarrow \mathit{uncountable}(\mathit{reals})}\;[\Rightarrow\text{ -intro}^{[1]}]$$

Cantor's diagonal argument (outline).

Example 12 : Constructive Proof Pattern

To constructively prove: $\exists x \bullet p(x)$

Strategy: 1. Explicitly construct a witness w 2. Verify $p(w)$ holds 3. Conclude $\exists x \bullet p(x)$ with $x = w$

Example: Prove $\exists n : \mathbb{N} \bullet n > 100 \wedge \mathit{even}(n)$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{true} \urcorner^{[1]}}{\mathit{witness\_construction}(102)}\;[\text{construction}]}{102 > 100}\;[\text{arithmetic}]}{102 = 2 * 51}\;[\text{arithmetic}]}{\mathit{even}(102)}\;[\text{definition}]}{102 > 100 \wedge \mathit{even}(102)}\;[\wedge\text{ -intro}]}{\exists n \bullet n > 100 \wedge \mathit{even}(n)}\;[\exists\text{ intro with n} = 102]}{\mathit{true} \Rightarrow (\exists n \bullet n > 100 \wedge \mathit{even}(n))}\;[\Rightarrow\text{ -intro}^{[1]}]$$

Example 13 : Proof Composition

Combine multiple proof techniques:

Theorem: Property P holds for all cases

Overall strategy: Case analysis + Induction + Contradiction

$$\dfrac{\dfrac{\dfrac{}{p(\mathit{base})}\;[\text{direct proof}] \qquad \dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \neg\, ((p(n)))\urcorner^{[2]}}{\mathit{contradiction\_derived}}\;[\text{proof steps}]}{\mathit{false}}\;[\text{contradiction}]}{p(n)}\;[\neg\text{ -intro}^{[2]}] \qquad \dfrac{\dfrac{\ulcorner p(n-1)\urcorner^{[3]}}{p(n)}\;[\text{inductive step}]}{p(n)}\;[\Rightarrow\text{ -intro}^{[3]}]}{p(n)}\;[\vee\text{ elim over subcases}]}{\forall\, n \bullet p(n)}\;[\vee\text{ elim over cases}]}{\mathit{true} \Rightarrow (\forall\, n \bullet p(n))}\;[\Rightarrow\text{ -intro}^{[1]}]$$

Example 14 : Best Practices for Complex Proofs

Guidelines for writing advanced proofs:

1. State strategy at the beginning 2. Label cases clearly 3. Discharge assumptions promptly 4. Reference lemmas explicitly 5. Show key algebraic steps 6. Justify non-obvious steps 7. Use proof by cases when structure suggests it 8. Use induction for recursive definitions 9. Use contradiction for negative conclusions 10. Verify base cases thoroughly

Example 15 : Proof Documentation

Document complex proofs:

- **Goal**: State what you're proving - **Strategy**: Explain the proof approach - **Lemmas needed**: List dependencies - **Key insights**: Highlight non-obvious steps - **Pitfalls**: Note where proof could go wrong - **Generalization**: Explain how proof extends

Well-documented proofs are maintainable and reusable.